Exercise (2.1.1).

Proof. Notice that each $f \in A$ with nonzero constant term is a unit. Indeed, after multiplying by an element of k, we may assume that the constant term is 1. I.e. $f = 1 + gT$ for some $g \in A$. Then, since A is T-adically complete, we have that

$$\sum_{k=0}^{\infty}(-gT)^k$$

converges to some $h \in A$. But then $hf = h(1 + gT) = 1$ directly, so f is invertible.

It is also clear that $\{0A, TA\} \subseteq \operatorname{Spec}(A)$, and I claim this list is complete. Indeed, suppose $P \in \operatorname{Spec}(A)$ is nonzero. Then it contains some nonzero element which can be factorized as $T^r f$ for some f with nonzero constant term. But we've shown that f is a unit, so P contains T^r, and since it's prime, it contains T. But TA is maximal, so $P = TA$ as desired. □

Exercise (2.1.2).

Proof. Let $\mathfrak{m} \in \operatorname{Spec}(B)$ be a closed point, i.e. a maximal ideal of B. Then we can take the composition:

$$A \xrightarrow{\varphi} B \to B/\mathfrak{m}$$

Then $B/\mathfrak{m}$ is a finitely generated k-algebra and a field, so it's a finite field extension (Zariski Lemma). So, each element of A maps to an algebraic element over k. I.e. the image of A in $B/\mathfrak{m}$ is a finitely generated k-algebra that is algebraic over k. Thus, it must be a field, so $A/\varphi^{-1}(\mathfrak{m})$ is a field, whence $\varphi^{-1}(\mathfrak{m})$ is maximal. So, $(\operatorname{Spec}\varphi)(\mathfrak{m}) = \varphi^{-1}(\mathfrak{m})$ is a closed point. □

Exercise (2.1.3).

Proof. Since $A/\mathfrak{m}$ is a finite field extension of $\mathbb{R}$, it is either $\mathbb{R}$ or $\mathbb{C}$. So, it injects into $\mathbb{C}$, giving:

$$\mathbb{R}[X, Y] \to A \to A/\mathfrak{m} \hookrightarrow \mathbb{C}$$

Then the images of $1, x, x^2$ in $\mathbb{C}$ are linearly dependent over $\mathbb{R}$, so there is some $a, b, c \in \mathbb{R}$ with

$$a + bx + cx^2 \in \mathfrak{m}$$

If c is nonzero, dividing by it gives the desired quadratic in x contained in $\mathfrak{m}$. Otherwise, we must have b nonzero, so dividing by b and squaring gives the desired quadratic. The same argument applies to y. The same argument applied to $1, x, y$ gives the final claimed relation:

$$f = \alpha x + \beta y + \gamma \in \mathfrak{m}$$

and if $(\alpha, \beta) = (0, 0)$, then $\gamma \in \mathfrak{m}$ and $\gamma \neq 0$, which contradicts properness of $\mathfrak{m}$.

We'd like to show $\mathfrak{m} = fA$. WLOG, assume $\beta \neq 0$. It suffices to show that $(f, X^2 + Y^2 + 1)$ is maximal in $\mathbb{R}[X, Y]$, where we also use f to denote $\alpha X + \beta Y + \gamma$. We'll take the quotient iteratively. First, we claim that $\mathbb{R}[X, Y]/(f) \cong \mathbb{R}[t]$. Indeed, consider the map $\psi : \mathbb{R}[X, Y] \to \mathbb{R}[t]$ given by:

$$X \mapsto t \text{ and } Y \mapsto -\frac{\alpha t + \gamma}{\beta}$$

Clearly $f \in \ker\psi$ and ψ surjects. Further, if $g \in \ker\psi$, then by polynomial division (in $\mathbb{R}[X][Y]$), we can write:

$$g(X, Y) = f(X, Y)q(X, Y) + h(X)$$

since β is a unit. Thus $\psi(h) = 0$, but $\psi(h) = h(t)$ is zero in $\mathbb{R}[t]$ iff it is zero in $\mathbb{R}[X]$. I.e. $g = fq \in (f)$. So, $\mathbb{R}[X, Y]/(f) \cong \mathbb{R}[t]$, and under this isomorphism, X^2+Y^2+1 maps to a quadratic $p(t)$ (it is actually quadratic since the leading coefficient is positive, hence nonzero).

So, finally, to show that (f, X^2+Y^2+1) is maximal, it suffices to show that p is irreducible, i.e. that it has no root. Suppose it has a root u. Then we have the composite:

$$v : \mathbb{R}[X, Y] \to \mathbb{R}[t] \to \mathbb{R}$$

where the final map is evaluation $t \mapsto u$. Under the composition, we have $X^2 + Y^2 + 1 \mapsto p(u) = 0$, but this would mean that $0 = v(X)^2 + v(Y)^2 + 1 \geq 1$, which is a contradiction. Hence p is irreducible and $(f, X^2 + Y^2 + 1)$ is maximal, so that

$\mathfrak{m} = fA$.

The above proof shows in general that given $\alpha, \beta, \gamma \in \mathbb{R}$ with $(\alpha, \beta) \neq (0, 0)$, that $(\alpha x + \beta y + \gamma)$ is maximal in A. Multiplying the vector (α, β, γ) does not change this ideal, so we indeed get a map $\mathbb{P}^2_{\mathbb{R}} \setminus \{(0:0:1)\} \to \operatorname{Spec}(A)$. We've also shown that this map surjects on the set of maximal ideals. So, we only need to show it injects. In other words, suppose $\mathfrak{m} = (\alpha x + \beta y + \gamma) = (\alpha' x + \beta' y + \gamma')$ for some $(\alpha : \beta : \gamma) \neq (\alpha' : \beta' : \gamma'] \in \mathbb{P}^2_{\mathbb{R}} \setminus \{[0:0:1]\}$. We have two cases. First, suppose (α, β) and (α', β') are independent in $\mathbb{R}^2$. Then they form a basis, so we can find $a, b, c, d \in \mathbb{R}$ with:

$$\begin{pmatrix} \alpha & \alpha' \\ \beta & \beta' \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Then $\mathfrak{m}$ contains:

$$a(\alpha x + \beta y + \gamma) + c(\alpha' x + \beta' y + \gamma') = x + r$$

for a constant $r \in \mathbb{R}$. Similarly, $\mathfrak{m}$ contains:

$$b(\alpha x + \beta y + \gamma) + d(\alpha' x + \beta' y + \gamma') = y + s$$

for a constant $s \in \mathbb{R}$. But then, in $A/\mathfrak{m}$ we have $x \mapsto -r$ and $y \mapsto -s$, which gives:

$$0 = x^2 + y^2 + 1 \mapsto r^2 + s^2 + 1 \geq 1$$

which is a contradiction.

So we must have that (α, β) and (α', β') are linearly dependent. Since they are nonzero, there is a constant c with $(\alpha, \beta) = c(\alpha', \beta')$. We cannot have $\gamma = c\gamma'$, since the two points are distinct in $\mathbb{P}^2_{\mathbb{R}}$. But then $\mathfrak{m}$ contains:

$$(\alpha x + \beta y + \gamma) - c(\alpha' x + \beta' y + \gamma') = \gamma - c\gamma'$$

which is nonzero, contradicting the fact that $\mathfrak{m}$ is proper. So, we've exhausted all possibilities and must have that the map is injective. Thus, the association $(\alpha : \beta : \gamma) \mapsto (\alpha x + \beta y + \gamma)A$ is a bijection between $\mathbb{P}^2_{\mathbb{R}} \setminus \{(0:0:1)\}$ and the maximal ideals of A.

As claimed, the map $\mathbb{R}[X] \to \mathbb{R}[X, Y] \to A$ is injective. Indeed, any element of the ideal $(X^2 + Y^2 + 1)$ necessarily has degree at least 2 in Y, while any polynomial in X has degree zero in Y. The morphism is also finite since we have a surjection of $\mathbb{R}[X]$-modules $\mathbb{R}[X]^2 \to A$ mapping (u, v) to $u + yv$. Indeed, given $g \in A$, we can lift it to $g(X, Y) \in \mathbb{R}[X, Y]$, and by polynomial division we can write

$$g(X, Y) = (X^2 + Y^2 + 1)q(X, Y) + Yr(X) + s(X)$$

so that after mapping into A we get:

$$g = yr(x) + s(x)$$

which is in the image of the $\mathbb{R}[X]$-module map above.

So now, if $\mathfrak{p}$ is a non-maximal prime of A, then I claim $\mathfrak{p} \cap \mathbb{R}[X]$ is a non-maximal prime of $\mathbb{R}[X]$. Indeed, let $\mathfrak{m}$ be a maximal ideal (strictly) containing $\mathfrak{p}$. Then, from the previous problem, $\mathbb{R}[X] \cap \mathfrak{m}$ is maximal and contains $\mathbb{R}[X] \cap \mathfrak{p}$. But we cannot have $\mathbb{R}[X] \cap \mathfrak{p} = \mathbb{R}[X] \cap \mathfrak{m}$ since A is an integral extension of $\mathbb{R}[X]$, which therefore satisfies incomparability. So $\mathbb{R}[X] \cap \mathfrak{p}$ is strictly contained in $\mathbb{R}[X] \cap \mathfrak{m}$, and therefore cannot be maximal.

But $\mathbb{R}[X]$ is a PID, so any nonzero prime is maximal, and therefore $\mathbb{R}[X] \cap \mathfrak{p} = 0$. Now let $g \in \mathfrak{p}$. Then $g \in A$, which is integral over $\mathbb{R}[X]$, so there is a relation:

$$g^n + a_{n-1}g^{n-1} + \cdots + a_0 = 0$$

where each $a_i \in \mathbb{R}[X]$. We may further assume that $n \geq 1$ is as small as possible. Now notice that a_0 is a multiple of g, so $a_0 \in \mathbb{R}[X] \cap \mathfrak{p} = 0$, so $a_0 = 0$. If $n > 1$, this gives

$$g^{n-1} + a_{n-1}g^{n-2} + \cdots + a_1 = 0$$

contradicting minimality of n. So instead we get $n = 1$, whence $g = -a_0 = 0$. So $\mathfrak{p} = 0$, and there is at most one non-maximal prime of A. $\square$

Exercise (2.1.4).

First, a geometric argument:

Proof. Notice that $\mathfrak{p}$ is an irreducible component of $\operatorname{Spec} A$. Then, if $f \in A_{\mathfrak{p}}$, then f is the germ of a function defined locally on $\mathfrak{p}$, and if $f \in \mathfrak{p}A_{\mathfrak{p}}$, then it vanishes on all of $\mathfrak{p}$. Hence, f vanishes everywhere, so by the Nullstellensatz, some power of it is zero.

Now, since A is reduced, only the zero function vanishes on all of $\operatorname{Spec} A$. Hence, if f is a zerodivisor, then there is some nonzero g with $fg = 0$, and g being nonzero implies that g does not vanish everywhere. Pick an irreducible component Z that g does not vanish on. Then

$$Z = Z \cap \operatorname{Spec}(A) = Z \cap V(fg) = Z \cap (V(f) \cup V(g)) = (Z \cap V(f)) \cup (Z \cap V(g))$$

By irreducibility, we must have $V(f) \supseteq Z$, so f vanishes on all of Z. □

Now, the formal algebra:

Proof. Note that $\mathfrak{p}A_{\mathfrak{p}}$ is the unique prime of $A_{\mathfrak{p}}$ since $\mathfrak{p}$ is minimal. Hence, the nilradical of $A_{\mathfrak{p}}$, as the intersection of the primes of $A_{\mathfrak{p}}$, is precisely $\mathfrak{p}A_{\mathfrak{p}}$. Hence, if $f \in \mathfrak{p}$ is nonzero, then its image is in $\mathfrak{p}A_{\mathfrak{p}}$, so there is some n such that $\frac{f^n}{1} = 0$. I.e. there is some $u \notin \mathfrak{p}$ with $f^n u = 0$, so f is a zerodivisor.

Now, let A be reduced, and let f be a zerodivisor, so there is some nonzero g with $fg = 0$. Then notice that g is not contained in every minimal prime, since otherwise it would be contained in

$$\bigcap_{\mathfrak{p} \text{ minimal prime}} \mathfrak{p} = \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p} = \sqrt{0} = 0$$

since A is reduced. So, there is some minimal prime $\mathfrak{p}$ with $g \notin \mathfrak{p}$. But $fg = 0 \in \mathfrak{p}$, so $f \in \mathfrak{p}$ by primality. □

Exercise (2.1.5).

Proof. Note that injectivity at the beginning and surjectivity at the end are obvious. So, we need to see exactness in the middle. Further, any multiple of $P_1(T_1)$ is clearly in the kernel, so the image is contained in the kernel. Finally, let $f \in k[T_1, \ldots, T_n]$ be in the kernel. Write it as a polynomial in $T_2, \ldots, T_n$ with coefficients in $k[T_1]$, as:

$$f = \sum_i a(T_1)T^i$$

where i runs over multi-indices and T^i denotes $T_2^{i_2} \cdots T_n^{i_n}$. Then, the image in $k_1[T_2, \ldots, T_n]$ is:

$$\sum_i \overline{a(T_1)}T^i$$

which must be zero. I.e. each coefficient must be zero, so $a(T_1) \in P_1(T_1)k[T_1]$. Thus, factoring out $P_1(T_1)$ from each summand shows that f is a multiple of $P_1(T_1)$.

Now, we induct on n. Indeed, $P_1(T_1) \subseteq \mathfrak{m}$, so $\mathfrak{m}$ corresponds to a maximal ideal $\mathfrak{m}_1$ of $k_1[T_2, \ldots, T_n]$. Let $p_2(T_2)$ generate $\mathfrak{m}_1 \cap k_1[T_2]$. Then, lift this back to a polynomial $P_2(T_1, T_2) \in k[T_1, T_2]$. Notice that if $f \in k[T_1, T_2] \cap \mathfrak{m}$, then the image is in $\mathfrak{m}_1 \cap k_1[T_2]$, so f is contained in (P_1, P_2). Let $k_2 = k_1[T_2]/(p_2)$ and continue in this way to get the desired sequence. □

Exercise (2.1.6).

Proof. Let Z be a closed subspace of a quasi-compact space X. Let $\{U_i\}$ be an open cover of Z. Then, each U_i is of the form $U_i' \cap Z$ for some U_i' open in X. Notice that $\{U_i'\} \cup \{X \setminus Z\}$ is an open cover of X. This has a finite subcover, say $U_1', \ldots, U_n', X \setminus Z$ (if this last set is not needed, it doesn't hurt to have it anyway). Then $U_1, \ldots, U_n$ is a cover of Z, so indeed any open cover of Z has a finite subcover. I.e. Z is quasi-compact.

Now, let $\{U_i\}$ be an open cover of $\operatorname{Spec} A$. By definition, each U_i is the complement of $V(J_i)$ for some ideal J_i of A. Then, we get:

$$V\left(\sum_i J_i\right) = \bigcap_i V(J_i) = \operatorname{Spec}(A) \setminus \left(\bigcup_i U_i\right) = \emptyset$$

I.e. this ideal is not contained in any prime ideal, and so cannot be proper. I.e.

$$\sum_i J_i = A$$

whence we have an equation:

$$\sum_i j_i = 1$$

where $j_i \in J_i$ and $j_i = 0$ for all but finitely many indices. But then if S is the support of this sum (the set of i where $j_i \neq 0$), then:

$$\sum_{i \in S} j_i = 1 \implies \sum_{i \in S} J_i = A$$

So:

$$\bigcup_{i \in S} U_i = \operatorname{Spec}(A) \setminus \bigcap_{i \in S} V(J_i) = \operatorname{Spec}(A) \setminus V\left(\sum_{i \in S} J_i\right) = \operatorname{Spec}(A) \setminus V(A) = \operatorname{Spec}(A)$$

so that we have found a finite subcover. I.e. $\operatorname{Spec} A$ is quasi-compact. □

Exercise (2.1.7).

Proof. It suffices to describe the irreducible closed subsets of $\operatorname{Spec} \mathbb{Z}[T]$, which are of the form $V(P)$ for $P \in \operatorname{Spec} \mathbb{Z}[T]$. Indeed, any closed set is then a finite union of these irreducible closed sets. We will organize this by $\operatorname{ht} P$:

The only height zero prime is (0), and $V(0) = \mathbb{A}^1_{\mathbb{Z}}$ is the whole space.

The height one primes are of the form (f) where $f \in \mathbb{Z}[T]$ is irreducible (over $\mathbb{Q}$, say), or of the form (p) for p a rational prime. I'm not sure what to say (geometrically) about $V(f)$, but note that $V(p) = \mathbb{A}^1_{\mathbb{F}_p}$.

Finally, the height two primes are of the form (f, p) for p a prime and f irreducible when reduced mod p (i.e. over $\mathbb{F}_p$). These are the closed points $V(f, p)$. □

Exercise (2.1.8).

Proof. In other words, we would like to show that if $\varphi : A \to B$ is integral and $\mathfrak{m} \in \operatorname{Spec} B$ is maximal, then $\varphi^{-1}\mathfrak{m}$ is maximal. Consider the induced map $\varphi A/\varphi^{-1}\mathfrak{m} \to B/\mathfrak{m}$. This is an injective integral extension (since the reduction of the same polynomials work) from a domain to a field. So, we must have that $A/\varphi^{-1}\mathfrak{m}$ is also a field, i.e. $\varphi^{-1}\mathfrak{m}$ is maximal as claimed.

The second claim is false. Indeed, if $\mathscr{O}$ is the ring of integers in a number field K and p is an unramified non-inert prime of the extension $\mathscr{O}/\mathbb{Z}$ lying over p, then the preimage $\operatorname{Spec}(\varphi)^{-1}(p\mathbb{Z})$ is the set of primes lying over p, which is not a singleton by choice. I.e. the preimage of a closed point is not a closed point (while $\mathbb{Z} \to \mathscr{O}$ is integral).

Directly, each element $b \otimes 1$ is integral over $A_{\mathfrak{p}}$ since it is integral over A and the same polynomial works. So, the integral closure of the image contains each simple tensor $b \otimes 1$ and each $1 \otimes (a/s)$, so it must be the full ring.

That T is multiplicative is obvious. To see that $T^{-1}B \neq 0$, it suffices to notice that $0 \notin T$ since φ is injective and $0 \notin A \setminus \mathfrak{p}$. So, lastly we show the isomorphism. Notice that we have an A-bilinear map $B \times A_{\mathfrak{p}} \to T^{-1}B$ given by:

$$\left(b, \frac{a}{s}\right) \mapsto \frac{\varphi(a)b}{\varphi(s)}$$

which thus factors through a map $f : B \otimes_A A_{\mathfrak{p}} \to T^{-1}B$. We also have a map $B \to B \otimes_A A_{\mathfrak{p}}$ given by $b \mapsto b \otimes 1$. If $t = \varphi(a) \in T$, where $a \notin \mathfrak{p}$, then it maps to $t \otimes 1$, which satisfies:

$$(t \otimes 1)\left(1 \otimes \frac{1}{a}\right) = \left(\varphi(a) \otimes \frac{1}{a}\right) = a \cdot \left(1 \otimes \frac{1}{a}\right) = 1 \cdot 1$$

so that each element of T maps to something invertible. So, the map factors through a map $g : T^{-1}B \to B \otimes_A A_{\mathfrak{p}}$. These are the desired isomorphisms. Indeed, for $b/t \in T^{-1}B$:

$$f(g(b/t)) = f((b \otimes 1)(t \otimes 1)^{-1}) = \frac{b}{1}\left(\frac{t}{1}\right)^{-1} = b/t$$

so this composition is the identity. For a simple tensor $b \otimes (a/s)$, we have:

$$g(f(b \otimes (a/s))) = g\left(\frac{\varphi(a)b}{\varphi(s)}\right) = (\varphi(a)b \otimes 1)(\varphi(s) \otimes 1)^{-1} = a(b \otimes 1)[s(1 \otimes 1)]^{-1} = a(b \otimes 1)(1 \otimes (1/s)) = b \otimes (a/s)$$

and so is identity on simple tensors, and hence on all of the domain. So, f and g do indeed establish the isomorphism.

This gives the final claim. Indeed, given $\mathfrak{p} \in \operatorname{Spec} A$, we have the square:

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ A_{\mathfrak{p}} \arrow[r] & B \otimes_A A_{\mathfrak{p}} = T^{-1}B \end{tikzcd}$$

Let $\mathfrak{m}$ be any closed point in $\operatorname{Spec} T^{-1}B$. Then, the image along the bottom arrow is a closed point in $\operatorname{Spec} A_{\mathfrak{p}}$, which therefore must be $\mathfrak{p}A_{\mathfrak{p}}$, as this is the unique closed point. Taking the further image along the localization map thus gives the ideal $\mathfrak{p} \in \operatorname{Spec} A$. But since this diagram commutes, the image in $\operatorname{Spec} B$ of $\mathfrak{m}$ gives a point of $\mathfrak{m}' \in \operatorname{Spec} B$ such that $\operatorname{Spec}(\varphi)(\mathfrak{m}') = \mathfrak{p}$ as desired. $\square$

Exercise (2.1.9).

Proof. First, note that since A is a finite k-module, the extension is integral. Let $\mathfrak{p} \in \operatorname{Spec} A$; then it is contained in some maximal ideal $\mathfrak{m} \in \operatorname{Spec} A$. Then $\mathfrak{p} \cap k$ and $\mathfrak{m} \cap k$ must be primes in k, i.e. they must both be zero. So, they lie over the same prime, and since integral extensions satisfy incomparability, the containment $\mathfrak{p} \subseteq \mathfrak{m}$ implies $\mathfrak{p} = \mathfrak{m}$. So, indeed every prime is maximal.

Now, suppose $\operatorname{Spec} A$ is has at least $d+1$ elements for $d = \dim_k(A)$. Then choose distinct primes (maximals) $\mathfrak{m}_1, \mathfrak{m}_2, \mathfrak{m}_3, \ldots, \mathfrak{m}_{d+1}$. Let

$$I_r = \bigcap_{j=1}^{r} \mathfrak{m}_j$$

Then we have the descending chain:

$$A \supseteq I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots I_{d+1}$$

Each of these containments must be proper. Indeed, otherwise we would have

$$\mathfrak{m}_1 \cdots \mathfrak{m}_r \subseteq I_r \subseteq \mathfrak{m}_{r+1}$$

But then, since $\mathfrak{m}_{r+1}$ is prime, it contains some $\mathfrak{m}_i$, whence $\mathfrak{m}_{r+1} = \mathfrak{m}_i$ by maximality, contrary to assumption.

On the other hand, each of these is a k-submodule of A, i.e. a k-vector space. Since the containments are proper, we must have that $\dim_k(I_{r+1}) < \dim_k(I_r)$. But $\dim_k(A) = d$, so this is a contradiction, as we cannot have a strictly decreasing sequence of length $d+2$ among the integers $\{0, \ldots, d\}$. Thus, we must have $|\operatorname{Spec} A| \leq \dim_k(A)$.

Note that $k[T_1, \ldots, T_d]$ is a UFD, and a nonzero polynomial is a unit iff it is a constant. Hence, Euclid's proof shows that there are infinitely many prime elements and so infinitely many primes. Explicitly: note that T_1 is prime, so there is at least one nonzero prime. Suppose that $\pi_1, \ldots, \pi_n$ is a list of nonzero primes, so that none of them may be constant. Then $a = \pi_1 \cdots pi_n + 1$ factors as a nonempty product of primes since it has positive degree. None of these primes can be any π_i, else π_i divides 1. So, no finite list of primes of positive degree is complete, hence there are infinitely many primes.

We've already shown one direction, so now suppose that A is not a finite k-module. By Noether normalization, we can find a finite injective morphism $k[T_1, \ldots, T_d] \to A$. We must have $d \geq 1$ since we've assumed A is not finite over k. But now, the extension A over $k[T_1, \ldots, T_d]$ satisfies lying over, so we have at least one prime of A over each prime of $k[T_1, \ldots, T_d]$. In fact, this association is injective since each prime of A lies over a unique prime of $k[T_1, \ldots, T_d]$. So, $|\operatorname{Spec} A| \geq |\operatorname{Spec} k[T_1, \ldots, T_d]| = \infty$ as desired. $\square$

Exercise (2.2.1).

Proof. Following the explicit construction given, the sheafification of the constant presheaf is the sheaf $\mathscr{A}^\dagger$ where $\mathscr{A}^\dagger(U)$ is the set of functions $f : U \to \bigsqcup_{x \in U} \mathscr{A}_x$ given locally by sections. Clearly $\mathscr{A}_x = A$, and the fact that f is locally given by sections means that for each $x \in U$, there is some open V and $a \in \mathscr{A}(V) = A$ with $f(x) = a_x = a$ for each $x \in V$. Another way to state this is that $f : U \to A$ is continuous when A is given the discrete topology. I.e. $\mathscr{A}^\dagger$ is the sheaf of continuous functions from subsets of X to A with the discrete topology, i.e. locally constant functions $X \to A$. Finally, we can interpret this as $\mathscr{A}^\dagger(U)$ being the direct product of copies of A in bijection with the connected components of U.

If every open subset of X is connected, then this description shows that $\mathscr{A} = \mathscr{A}^\dagger$, so $\mathscr{A}$ is a sheaf. Conversely, if there is some $U \subseteq X$ that is the disjoint union of open subsets V, W of X, then $\mathscr{A}$ is not a sheaf. Indeed, A is nontrivial, so we can choose two distinct elements $a, b \in A$. Then $a \in \mathscr{A}(V)$ and $b \in \mathscr{A}(W)$, and these sections agree on overlaps since $V \cap W = \emptyset$, but there is no $x \in \mathscr{A}(U)$ with $x|_V = x = a$ and $x|_W = x = b$. So, $\mathscr{A}$ does not satisfy gluing. $\square$

Exercise (2.2.2).

Proof. Let $A = \{x \in X : s_x = t_x\}$. Suppose that $x \in A$. Then there is an open neighborhood U of x with $s|_U = t|_U$. But then for any $y \in U$, we have

$$s_y = (s|_U)_y = (t|_U)_y = t_y$$

so $y \in A$. Thus $U \subseteq A$ and we've shown that each $x \in A$ is contained in an open set contained in A, so A is open as claimed. □

Exercise (2.2.3).

Proof. I'll come back to this one. □

Exercise (2.2.4).

Proof. Name the maps:

$$0 \to \mathscr{F}' \xrightarrow{\varphi} \mathscr{F} \xrightarrow{\psi} \mathscr{F}'' \to 0$$

Recall that this implies that for each $x \in X$ that we have an exact sequence:

$$0 \to \mathscr{F}'_x \xrightarrow{\varphi_x} \mathscr{F}_x \xrightarrow{\psi_x} \mathscr{F}''_x \to 0$$

Now, exactness at $\mathscr{F}'(X)$ is obvious, since a map of sheaves is injective iff it is injective on each open set. So, we would like to show exactness at $\mathscr{F}(X)$. First, let $s \in \mathscr{F}'(X)$. Then for each $x \in X$ we have:

$$[\psi(X)(\varphi(X)(s))]_x = \psi_x([\varphi(X)(s)]_x) = \psi_x(\varphi_x(s_x)) = 0$$

since $\psi_x \circ \varphi_x = 0$. Hence the image of $\varphi(X)$ is contained in the kernel of $\psi(X)$. Conversely, suppose $s \in \ker(\psi(X)) \subseteq \mathscr{F}(X)$. Then, again, for each x, we have

$$\psi_x(s_x) = [\psi(X)(s)]_x = 0$$

So, $s_x \in \ker(\psi_x) = \operatorname{im}(\varphi_x)$. Hence, for each x we can find an open neighborhood $U(x)$ of x and a section $t(x) \in \mathscr{F}'(U(x))$ with $\varphi(U(x))(t(x)) = s|_{U(x)}$. We would like to glue these to a single global section, and so we need to see that they agree on overlaps.

To that end, let $x, y \in X$, let $a = t(x)|_{U(x)\cap U(y)}$, and let $b = t(y)|_{U(x)\cap U(y)}$. If $U(x) \cap U(y) = \emptyset$, then we must have $a = b$, and there is nothing to show. Otherwise, let $z \in U(x) \cap U(y)$. Then we have:

$$\varphi_z(a_z) = \varphi_z(t(x)_z) = [\varphi(U(x))(t(x))]_z = (s|_{U(x)})_z = s_z$$

and similarly:

$$\varphi_z(b_z) = \varphi_z(t(y)_z) = [\varphi(U(y))(t(y))]_z = (s|_{U(y)})_z = s_z$$

So, $\varphi_z(a_z) = \varphi_z(b_z)$ and by injectivity of φ_z we get $a_z = b_z$. Since this is true for all $z \in U(x) \cap U(y)$, we get that $a = b$, and so the $t(x)$ do indeed agree on overlaps. Hence they glue to a single section $t \in \mathscr{F}'(X)$ with $t|_{U(x)} = t(x)$.

But then we are finished, for if $x \in X$, then:

$$[\varphi(X)(t)]_x = \varphi_x(t_x) = \varphi_x((t|_{U(x)})_x) = \varphi_x(t(x)_x) = [\varphi(U(x))(t(x))]_x = (s|_{U(x)})_x = s_x$$

and so $\varphi(X)(t) = s$ since they are equal on all stalks. I.e. s is in the image of $\varphi(X)$. □

Exercise (2.2.5).

Proof. □

Exercise (2.2.6).

Proof. □

Exercise (2.2.7).

Proof. □

Exercise (2.2.8).

Proof. □

Exercise (2.2.9).

Proof. □

Exercise (2.2.10).

Proof. □

Exercise (2.2.11).

Proof. □

Exercise (2.2.12).

Proof. □

Exercise (2.2.13).

Proof. □

Exercise (2.2.14).

Proof. □

Exercise (2.3.1).

Proof. □

Exercise (2.3.2).

Proof. Let $i : U \to X$ be the inclusion map, so that we wish to show $i^\# : A \to B$ is flat. It suffices to show this locally. That is, for $\mathfrak{q} \in \operatorname{Spec} B = U$ and $\mathfrak{p} = i(\mathfrak{q}) \in \operatorname{Spec} A$, we wish to show that $A_\mathfrak{p} \to B_\mathfrak{q}$ is flat. But this is the map $i^\#_\mathfrak{q} : U_\mathfrak{q} \to X_{i(\mathfrak{q})}$, which is an isomorphism since U is an open subscheme. Isomorphisms are clearly flat, so the claim is shown. □

Exercise (2.3.3).

Proof. Notice that if $\mathfrak{p} \in F$, then $\mathfrak{p} \supseteq I$ tautologically, and so $\mathfrak{p} \in V(I)$ by definition. I.e. we immediately have $F \subseteq V(I)$, and since $V(I)$ is closed, this gives $\overline{F} \subseteq V(I)$. We wish to show the reverse.

Conversely, we have that $\overline{F}$ is a closed set, so it is of the form $V(J)$ for some ideal J. Then:

$$J \subseteq \sqrt{J} = \bigcap_{\mathfrak{p} \in V(J)} \mathfrak{p} = \bigcap_{\mathfrak{p} \in \overline{F}} \mathfrak{p} \subseteq \bigcap_{\mathfrak{p} \in F} \mathfrak{p} = I$$

where the inclusion comes from the fact that we are intersecting over a subset. So, by applying V, we get: $\overline{F} = V(J) \supseteq V(I)$, as desired.

Notice that for $\varphi : A \to B$ and $f : \operatorname{Spec} B \to \operatorname{Spec} A$, $\mathfrak{p} \in \operatorname{im} f \iff \mathfrak{p} \supseteq \ker \varphi \iff \mathfrak{p} \in V(\ker \varphi)$. So:

$$\overline{\operatorname{im} f} = V\left(\bigcap_{\mathfrak{p} \in \operatorname{im} f} \mathfrak{p}\right) = V\left(\bigcap_{\mathfrak{p} \in V(\ker \varphi)} \mathfrak{p}\right) = V(\sqrt{\ker \varphi}) = V(\ker \varphi)$$

as claimed. □

Exercise (2.3.4).

Proof. □

Exercise (2.3.5).

Proof. □

Exercise (2.3.6).

Proof. □

Exercise (2.3.7).

Proof. □

Exercise (2.3.8).

Proof. □

Exercise (2.3.9).

Proof. □

Exercise (2.3.10).

Proof. □

Exercise (2.3.11).

Proof. □

Exercise (2.3.12).

Proof. □

Exercise (2.3.13).

Proof. □

Exercise (2.3.14).

Proof. □

Exercise (2.3.15).

Proof. □

Exercise (2.3.16).

Proof. □

Exercise (2.3.17).

Proof. □

Exercise (2.3.18).

Proof. □

Exercise (2.3.19).

Proof. □

Exercise (2.3.20).

Proof. □

Exercise (2.3.21).

Proof. □

Exercise (2.3.22).

Proof. □

Exercise (2.4.1).

Proof. □

Exercise (2.4.2).

Proof. □

Exercise (2.4.3).

Proof. □

Exercise (2.4.4).

Proof. □

Exercise (2.4.5).

Proof. □

Exercise (2.4.6).

Proof. □

Exercise (2.4.7).

Proof. □

Exercise (2.4.8).

Proof. □

Exercise (2.4.9).

Proof. □

Exercise (2.4.10).

Proof. □

Exercise (2.4.11).

Proof. □

Exercise (2.4.12).

Proof. □

Exercise (2.5.1).

Proof. □

Exercise (2.5.2).

Proof. □

Exercise (2.5.3).

Proof. □

Exercise (2.5.4).

Proof. □

Exercise (2.5.5).

Proof. □

Exercise (2.5.6).

Proof. □

Exercise (2.5.7).

Proof. □

Exercise (2.5.8).

Proof. □

Exercise (2.5.9).

Proof. □

Exercise (2.5.10).

Proof. □

Exercise (2.5.11).

Proof. □

Exercise (2.5.12).

Proof. □

Exercise (2.5.13).

Proof. □

Exercise (2.5.14).

Proof. □

Exercise (2.5.15).

Proof. □